Algebra Review: Unit 5 — Systems of Equations

Station 1 of 4: Checking Solutions and Types of Systems

Name: _____ Date: _____

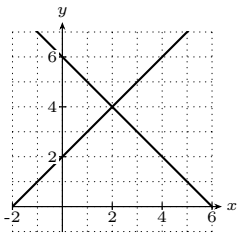
Substitute to check solutions. For the remaining problems, find the intersection or describe the number of solutions. Diagrams are not drawn to scale.

1. Is $(3, 1)$ a solution of the system? $\begin{cases} x + y = 4 \\ 2x - y = 5 \end{cases}$

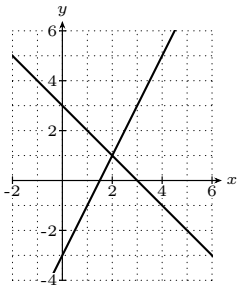
2. Is $(1, 2)$ a solution of the system? $\begin{cases} y = x + 1 \\ y = -x + 4 \end{cases}$

3. Is $(5, -1)$ a solution of the system? $\begin{cases} 3x + 2y = 13 \\ x - y = 6 \end{cases}$

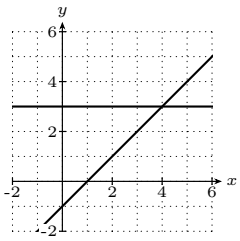
4. Use the graph to find the solution of the system: $\begin{cases} y = x + 2 \\ y = -x + 6 \end{cases}$



5. Use the graph to find the solution of the system: $\begin{cases} y = 2x - 3 \\ y = -x + 3 \end{cases}$



6. Use the graph to find the solution of the system: $\begin{cases} y = 3 \\ y = x - 1 \end{cases}$



7. How many solutions does the system have? $\begin{cases} y = 2x + 3 \\ y = 2x - 5 \end{cases}$

8. How many solutions does the system have? $\begin{cases} x + 2y = 4 \\ 3x + 6y = 12 \end{cases}$

9. Without graphing, how many solutions does $\begin{cases} y = 3x - 2 \\ y = -x + 6 \end{cases}$ have? Then find it.

10. For what value of k does $\begin{cases} y = 3x + 1 \\ y = kx - 4 \end{cases}$ have no solution?

Algebra Review: Unit 5 — Systems of Equations

Station 2 of 4: Solving Systems by Substitution

Name: _____ Date: _____

Solve each system by substitution. Write your answer as an ordered pair (x, y).

1. Solve the system: $\begin{cases} y = 2x \\ x + y = 12 \end{cases}$

2. Solve the system: $\begin{cases} y = x - 3 \\ 2x + y = 9 \end{cases}$

3. Solve the system: $\begin{cases} x = 3y + 1 \\ 2x - y = 7 \end{cases}$

4. Solve the system: $\begin{cases} y = -2x + 5 \\ 3x + y = 7 \end{cases}$

5. Solve the system: $\begin{cases} x = 5 - y \\ 3x + 2y = 12 \end{cases}$

6. Solve the system: $\begin{cases} 2x + y = 11 \\ y = x + 2 \end{cases}$

7. Solve the system: $\begin{cases} x + y = 10 \\ y = 4x \end{cases}$

8. Solve the system: $\begin{cases} x + 2y = 7 \\ 3x - y = 7 \end{cases}$

9. Solve the system: $\begin{cases} 4x - y = 5 \\ 2x + 3y = 13 \end{cases}$

10. Two numbers have a sum of 25. One number is 7 more than the other. Find the numbers.

Algebra Review: Unit 5 — Systems of Equations

Station 3 of 4: Solving Systems by Elimination

Name: _____ Date: _____

Solve each system by elimination. Write your answer as an ordered pair (x, y) or state the number of solutions.

1. Solve the system: $\begin{cases} x + y = 9 \\ x - y = 3 \end{cases}$

2. Solve the system: $\begin{cases} 2x + y = 11 \\ 3x - y = 4 \end{cases}$

3. Solve the system: $\begin{cases} 3x + 2y = 16 \\ x - 2y = 0 \end{cases}$

4. Solve the system: $\begin{cases} 5x + 3y = 19 \\ 5x - 2y = 4 \end{cases}$

5. Solve the system: $\begin{cases} 2x + 3y = 12 \\ 4x - y = 10 \end{cases}$

6. Solve the system: $\begin{cases} 3x + 4y = 5 \\ 2x + 3y = 4 \end{cases}$

7. Solve the system: $\begin{cases} 2x - 5y = 1 \\ 3x + 2y = 11 \end{cases}$

8. Solve the system: $\begin{cases} 7x - 2y = 3 \\ -7x + 5y = 3 \end{cases}$

9. Solve: $\begin{cases} 2x + 3y = 6 \\ 4x + 6y = 18 \end{cases}$

10. Adult tickets cost \$8 and student tickets cost \$5. A total of 40 tickets were sold for \$245. How many of each were sold?

Algebra Review: Unit 5 — Systems of Equations

Station 4 of 4: Choosing a Method and Systems in Context

Name: _____ Date: _____

Write a system for each situation and solve it. For the last problems, identify how many solutions the system has.

1. Gym A charges a \$10 fee plus \$4 per visit. Gym B charges a \$30 fee plus \$2 per visit. After how many visits v do both cost the same, and what is that cost?

2. A jar has 30 coins, all nickels and dimes, worth \$2.25. How many nickels and how many dimes?

3. A boat travels 24 miles downstream in 2 hours and returns upstream in 3 hours. Find the speed of the boat in still water and the speed of the current.

4. The sum of two numbers is 18 and their difference is 4. Find the numbers.

5. The perimeter of a rectangle is 30 cm. The length is 3 cm more than the width. Find the dimensions.

6. A number is 4 times another number. Their sum is 35. Find the numbers.

7. Five notebooks and three pens cost \$19. Two notebooks and four pens cost \$16. Find the price of one notebook and one pen.

8. Choose a method and solve: $\begin{cases} 3x + y = 7 \\ x - 2y = 0 \end{cases}$

9. How many solutions? $\begin{cases} y = 4x - 1 \\ 8x - 2y = 2 \end{cases}$

10. How many solutions? $\begin{cases} y = -x + 5 \\ 2x + 2y = 7 \end{cases}$

Algebra Review: Unit 5 — Systems of Equations

Answer Key

Station 1: Checking Solutions and Types of Systems

1. Yes
2. No
3. Yes
4. $(2,\ 4)$
5. $(2,\ 1)$
6. $(4,\ 3)$
7. No solution *same slope, different y-intercepts (parallel lines)*
8. Infinitely many solutions *the second equation is 3 times the first (same line)*
9. Exactly one: $(2, 4)$ *slopes differ*
10. $k = 3$ *parallel lines need equal slopes*

Station 2: Solving Systems by Substitution

1. $(4,\ 8)$
2. $(4,\ 1)$
3. $(4,\ 1)$
4. $(2,\ 1)$
5. $(2,\ 3)$
6. $(3,\ 5)$
7. $(2,\ 8)$
8. $(3,\ 2)$
9. $(2,\ 3)$
10. 9 and 16 $x + y = 25,\ x = y + 7$

Station 3: Solving Systems by Elimination

1. $(6,\ 3)$
2. $(3,\ 5)$
3. $(4,\ 2)$
4. $(2,\ 3)$
5. $(3,\ 2)$
6. $(-1,\ 2)$
7. $(3,\ 1)$
8. $(1,\ 2)$
9. No solution *multiplying the first equation by 2 gives* $4x + 6y = 12 \neq 18$
10. 15 adult and 25 student tickets $a+s = 40,\ 8a+5s = 245$

Station 4: Choosing a Method and Systems in Context

1. $10 + 4v = 30 + 2v;\ v = 10$ visits, cost \$50
2. 15 nickels and 15 dimes $n + d = 30,\ 5n + 10d = 225$
3. boat 10 mph, current 2 mph $b + c = 12,\ b - c = 8$
4. 11 and 7
5. width 6 cm, length 9 cm
6. 7 and 28
7. notebook \$2, pen \$3 $5n + 3p = 19,\ 2n + 4p = 16$
8. $(2,\ 1)$
9. Infinitely many *substitution gives* $2 = 2$
10. No solution *substitution gives* $10 = 7$*, which is false*